

Stanley Chukwuebuka Okonkwo

MSc, Ecology and Nature Management (RUDN University, 2026)

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Motivational Letter

I am applying for the PhD programme in Life Sciences at the Skolkovo Institute of Science and Technology, with prospective supervision by Professor Raghavendra Belur Jana at the Digital Agriculture Lab and Centre for AgroTechnologies. Professor Jana's research on remote sensing-based monitoring of hydrological and agricultural processes, integrated with AI and machine learning for environmental prediction, is directly aligned with the work I have been developing in my MSc research and subsequent projects.

I completed an MSc in Ecology and Nature Management at RUDN University in Moscow in June 2026. My thesis examined the ecological impacts of urbanisation on green spaces and drainage patterns in Lagos State, Nigeria, using multi-temporal Landsat 8/9 and Sentinel-2 satellite imagery processed in Google Earth Engine and ArcGIS. The work combined NDVI time-series analysis, land surface temperature mapping, and spatial modelling of drainage network disruption at the state level. That research made it clear that the most pressing environmental problems affecting agricultural and peri-urban communities in West Africa are fundamentally problems of land and water: how land conversion alters the hydrological processes that agricultural systems depend on, and how remote sensing and computational modelling can be used to understand and predict those changes before they become irreversible.

Professor Jana's work on applying AI and machine learning to agro-hydrological modelling, optimising water use under climate change, and integrating remote sensing with process-based environmental models is the methodological framework I want to develop further in doctoral research. The specific research direction I want to pursue sits at the intersection of his expertise and my existing dataset: using machine learning approaches to model the relationship between land cover change in rapidly urbanising West African landscapes and the agricultural water stress and soil degradation that follows, to produce spatially explicit predictions that can guide land use planning and agricultural adaptation at the community scale.

Beyond my thesis research, I have developed practical skills in Python-based data processing, built EcoSim (a web-based ecological simulation platform for West African biomes that integrates real-time soil moisture, temperature, and biodiversity data overlays), and contributed to community engagement programmes on sustainable land management and environmental monitoring in southeastern Nigeria. I am proficient in GIS platforms including ArcGIS and QGIS, Google Earth Engine, and scientific data analysis workflows. Skoltech's location in Moscow, where I am currently based, makes this an immediate and practical step. I have two peer-reviewed publications in environmental and geoscience research and am available to begin immediately.

Yours sincerely,

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